Hybrid Fuzzy-NSGA-II Intelligent Fleet Dispatcher

Multi-Objective Vehicle Routing with Fuzzy Time Windows, Mamdani Traffic Inference & 5-Algorithm Benchmarking

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Abstract

Last-mile urban freight logistics represents between 40% and 50% of aggregate supply chain expenditures and drives disproportionate carbon emissions in rapidly urbanizing corridors. The underlying Vehicle Routing Problem with Time Windows (VRPTW) is a classic NP-Hard combinatorial optimization challenge, yielding approximately 1.55 × 10^25 candidate routes for a moderate 25-node distribution network. Classical hard-computing methodologies rely on crisp, binary time windows that penalize a single second of arrival delay as a complete 100% service-level failure (0 utility), disregarding human tolerance, weather hazards, and urban traffic volatility. Furthermore, conventional formulations optimize pure geometric transit distance rather than true physical energy consumption governed by instantaneous vehicle payload.

This research presents **FleetPulse AI**, an integrated soft-computing framework combining continuous Fuzzy Set Theory, a 9-Rule Mamdani Environmental Inference Matrix, and a Memetic Non-Dominated Sorting Genetic Algorithm II (Hybrid Fuzzy-NSGA-II). The mathematical model simultaneously optimizes two conflicting objectives under a strict capacity constraint of ≤ 100 kg/van:

1. **Minimizing Load-Dependent Fleet Energy Cost (f_1)**, incorporating tare-to-gross payload friction physics;
2. **Maximizing Priority-Weighted Fuzzy Customer Satisfaction (f_2)**, parameterized via multi-kernel membership functions (Trapezoidal, Gaussian, Sigmoidal).

We benchmark the proposed Hybrid Fuzzy-NSGA-II against four competing paradigms: Multi-Objective Particle Swarm Optimization (MOPSO), Multi-Objective Ant Colony Optimization (MO-ACO), Standard Single-Objective Genetic Algorithm (SGA), and a Greedy Nearest-Neighbor Heuristic. Tested on real-world GIS coordinates from the Amaravati–Vijayawada metropolitan corridor and standardized Solomon VRPTW benchmarks (C101 clustered and R101 random), Hybrid Fuzzy-NSGA-II consistently dominates the Pareto landscape. On the 18-node Amaravati corridor under moderate traffic (1.18×), the proposed framework secures **Rank #1** with **73.50% Fuzzy Customer Satisfaction**, an **Energy Cost of 374.70**, and a dominant **Hypervolume (HV) of 21.96**, outperforming standalone MO-ACO (HV 21.23), MOPSO (HV 14.81), and SGA (HV 11.75). A 5-seed Monte Carlo statistical validation confirms deterministic Pareto archive stability with 0.00 standard deviation in energy and satisfaction. Finally, we formulate a commercial SaaS deployment roadmap for quick-commerce dark stores and emergency pharmaceutical distribution.

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1. Introduction

1.1 Background of Urban Logistics & Last-Mile Delivery Challenges

Urban logistics and last-mile freight transportation serve as the primary operational backbone of contemporary e-commerce, pharmaceutical supply chains, and rapid-delivery grocery networks. However, empirical transportation economic studies demonstrate that **last-mile delivery accounts for 40% to 50% of total supply chain logistics costs** and generates a disproportionately high share of urban greenhouse gas emissions.

In an operational deployment, fleet dispatch managers must constantly reconcile diametrically opposed objectives:

- **Vehicle Energy & Operating Cost:** Minimizing the fuel or electrical energy consumed by a heterogeneous delivery fleet. Fleet operating costs are not merely a linear function of distance; they depend heavily on the vehicle's dynamic physical weight, stopping frequency, and congested urban road segments.
- **Physical Vehicle Payload Limits:** Adhering strictly to payload constraints (max payload ≤ 100 kg/van) to avoid mechanical axle wear, regulatory fines, and battery exhaustion.
- **Strict Service Level Agreements (SLAs):** Delivering high-priority parcels within customer-stipulated time windows. Failing to satisfy critical delivery deadlines (e.g., emergency medical logistics, blood bank samples, airport air-cargo handoffs) incurs steep reputational damage and contractual penalties.

1.2 The NP-Hard Combinatorial Routing Challenge

The Vehicle Routing Problem with Time Windows (VRPTW) is a well-established **NP-Hard combinatorial optimization problem**. The problem belongs to the class of permutation-based routing problems where the candidate solution space scales factorially with the number of delivery nodes N:

$$\Omega(N) = (N-1)!2$$

For a moderate urban logistics deployment of N = 25 active delivery nodes, the number of unconstrained permutations is:

$$25! \approx 1.55 \times 10^{25} \text{ routes}$$

Evaluating this search space via brute-force enumeration at a processing speed of 10^9 evaluations per second would require over **4.9 × 10^8 years**, rendering deterministic exhaustive search computationally infeasible.

1.3 Limitations of Crisp / Hard Computing Formulations

Classical hard-computing approaches model customer deadlines as **crisp binary intervals**:

$$\mu_{crisp}(t) = \begin{cases} 1, & \text{if } t \in [T_{start}, T_{end}] \\ 0, & \text{otherwise} \end{cases}$$

Under this binary regime, an arrival that occurs 1 second past T_end is categorized as a complete 100% SLA failure (0 utility), identical to a parcel delivered five hours late. This mathematical rigidity fails to mirror realistic human behavioral tolerance and cannot adapt to urban stochastic uncertainties such as sudden traffic snarls, monsoonal downpours, or brief dispatch delays.

1.4 The Soft Computing Paradigm: Fuzzy Logic + Evolutionary Metaheuristics

Soft computing provides the mathematical tools required to conquer this dual challenge:

1. **Fuzzy Logic (Zadeh, 1965):** Provides continuous membership functions μ(t) in [0.0, 1.0] that capture human tolerance gradients during arrival delays, transforming brittle step-penalties into differentiable or smooth fitness surfaces.
2. **Mamdani Fuzzy Inference System:** Implements linguistic rule-based reasoning ("IF Traffic is Peak AND Weather is Monsoon THEN Delay is Gridlock") to compute realistic travel multipliers without requiring intractable differential equations of metropolitan traffic flows.
3. **Multi-Objective Evolutionary Computation (NSGA-II):** Simultaneously discovers non-dominated trade-offs between fleet energy cost (f_1) and customer satisfaction (f_2) using fast non-dominated sorting and crowding distance preservation.

1.5 Problem Statement & Research Questions

This project investigates the following primary research questions:

- *RQ1:* Can continuous fuzzy time windows eliminate the combinatorial brittleness of crisp VRPTW models while reflecting human tolerance?
- *RQ2:* How significantly does load-dependent energy dissipation alter the optimal sequencing of deliveries compared to standard geographic distance minimization?
- *RQ3:* Does a Memetic Hybrid Fuzzy-NSGA-II outperform alternative swarm (MOPSO) and colony (MO-ACO) paradigms across real-world GIS corridors and standard Solomon benchmarks?

2. Literature Review

2.1 Recent Academic Base Papers Referenced (2024–2026)

This project directly grounds its mathematical formulations, physical constraints, and algorithmic designs on three premier academic base papers published between 2024 and 2026:

- **Base Paper 1 (Dec 2024 - MDPI Systems, Vol. 12, Issue 12):**

Title: "Modeling and Solving the Multi-Objective Vehicle Routing Problem with Soft and Fuzzy Time Windows" Contribution Adopted: Formalized the mathematical foundations of continuous fuzzy membership functions to model customer delivery satisfaction. Demonstrated that modeling tolerance intervals [best_end, max_late] via linear and non-linear membership kernels eliminates the combinatorial brittleness of crisp VRPTW models and provides a continuous gradient for evolutionary search.

- **Base Paper 2 (2025 - IEEE International Conference on Intelligent Transportation Systems - ITSC):**

Title: "ACO-NSGAII: A Novel Metaheuristics for Bi-Objective Electric Vehicle Routing Problems with Time Windows" Contribution Adopted: Introduced an authentic load-dependent energy dissipation physics model where the energy consumed per kilometer scales linearly with instantaneous remaining vehicle payload. Established that prioritizing earlier drops for heavier parcels yields substantial cumulative battery savings.

- **Reference Paper 3 (Oct 2026 - Urban Science):**

Title: "The Prediction Optimization Bridge (POB): A Surrogate Assisted Random Forest–NSGA-II Framework for Energy Efficient Vehicle Routing" Contribution Adopted: Formulated real-time urban traffic delay multipliers that dynamically scale vehicle transit velocity and energy burn under varying weather and peak congestion conditions.

2.2 Comparative Literature Survey Matrix

Paper Citation & Year	Modeling Paradigm	Optimization Objectives	Environmental & Traffic Logic	Metaheuristic Solvers	Benchmark Datasets Tested	Key Limitations
MDPI Systems (Dec 2024)	Fuzzy Time Windows	Distance + Satisfaction	Static travel times; no weather inference	Standalone NSGA-II	Solomon C101, R101	Evaluates single algorithm; lacks load-dependent energy physics.
IEEE ITSC (2025)	Crisp VRPTW	Energy Cost + Fleet Size	Deterministic speed profiles	ACO-NSGA-II Hybrid	Synthetic 20-node EV networks	Binary time windows; zero fuzzy human tolerance modeling.
Urban Science (Oct 2026)	Surrogate NSGA-II	Carbon Emissions + Travel Time	Machine learning delay multipliers	Surrogate-assisted NSGA-II	European arterial highways	High training overhead; no swarm/colony comparative benchmarking.
FleetPulse AI (This Project, 2026)	**Continuous Fuzzy Set Theory + Mamdani Inference**	**Bi-Objective: Load-Dependent Energy (f_1) + Fuzzy Satisfaction (f_2)**	**9-Rule Mamdani Matrix (Traffic Density + Monsoon Weather)**	**5-Algorithm Arena: Hybrid Fuzzy-NSGA-II, MOPSO, MO-ACO, SGA, Greedy**	**Amaravati–Vijayawada Live GIS (25 Nodes) + Solomon C101 & R101**	**Unified multi-paradigm benchmark; sub-second runtime (<450 ms).**

2.3 Critical Analysis of Prior Works & Identified Research Gaps

A rigorous review of contemporary routing optimization literature reveals three prominent research gaps:

1. **Lack of Multi-Paradigm Soft Computing Benchmarking:** Existing studies evaluate a single isolated evolutionary algorithm (e.g., standalone NSGA-II or GA) without rigorous head-to-head benchmarking against alternative nature-inspired swarm (MOPSO) and colony (MO-ACO) paradigms under identical constraints, objective formulations,

and datasets.

2. **Absence of Multi-Kernel Fuzzy Satisfaction Comparisons:** Prior publications adopt a single arbitrary membership function (typically standard trapezoidal) without mathematically investigating how Gaussian (smooth bell-curve) or Sigmoidal (logistic S-curve) kernels influence evolutionary convergence and schedule structure.
3. **Decoupled Traffic and Energy Models:** Most classical studies optimize pure geographic distance, ignoring the multiplicative interaction between instantaneous cargo load and urban traffic delay factors.

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3. Motivation

3.1 Economic Impact: Inefficiencies in Urban Freight Delivery

Urban logistics is experiencing unprecedented growth driven by quick-commerce platforms (Zepto, Blinkit, Instamart) and omnichannel retail. However, last-mile delivery remains the most expensive and inefficient leg of the logistics chain, consuming **40% to 50% of aggregate supply chain expenditures**. Route inefficiency, partial vehicle capacity utilization, and unnecessary backtracking directly inflate vehicle fuel and fleet maintenance bills.

3.2 Environmental & Carbon Emission Pressures

Transportation is a primary contributor to urban air pollution (PM_2.5, NO_x) and greenhouse gas emissions. Most commercial delivery vehicles carry heterogeneous parcel weights that dramatically alter fuel efficiency or battery depletion. Minimizing pure geometric distance fails to minimize carbon footprint: traveling 10 km with a fully loaded van burns substantially more fuel than traveling the same distance with an empty van. A physics-grounded dispatch model is therefore an urgent environmental necessity.

3.3 Realistic SLA Modeling & Human Psychological Tolerance

In real-world logistics, customer satisfaction does not plummet to zero the moment a delivery is one minute late. Human psychology exhibits a tolerance band where utility degrades gradually before reaching an unacceptable threshold. Crisp binary deadlines force optimization algorithms to reject feasible, highly cost-effective routes due to trivial 30-second delays. Modeling satisfaction through continuous fuzzy membership functions restores realistic trade-off analysis.

3.4 Stochastic Urban Traffic & Monsoonal Dynamics

Metropolitan transport networks in South Asia (such as the Vijayawada–Amaravati highway corridor) suffer from high variability due to peak office hours and seasonal monsoon downpours. Hard-coded transit speeds lead to catastrophic schedule cascading delays. An adaptive soft-computing layer capable of inferring traffic delays from linguistic environmental conditions is essential.

3.5 Multi-Paradigm Soft Computing Benchmarking Imperative

Engineering decisions in enterprise logistics require empirical proof of which metaheuristic paradigm performs best under capacity and SLA constraints. By benchmarking Genetic Algorithms (NSGA-II, SGA), Particle Swarms (MOPSO), and Ant Colonies (MO-ACO) side-by-side, this study provides actionable mathematical and empirical clarity.

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4. Proposed Work

4.1 Core Engineering & Research Contributions

The primary contributions of this project are:

1. **Bi-Objective Mathematical Formulation:** Merging continuous multi-kernel fuzzy satisfaction with dynamic load-dependent energy physics under a $a \leq 100$ kg/van payload constraint.
2. **Tri-Kernel Fuzzy Satisfaction Engine:** Mathematical formulation and empirical evaluation of Trapezoidal, Gaussian, and Sigmoidal membership functions.
3. **9-Rule Mamdani Environmental Matrix:** Automated inference of speed and energy multipliers from traffic density and monsoon weather states.
4. **Memetic Hybrid Fuzzy-NSGA-II Solver:** Integrating heuristic chromosome seeding (EDF + Spatial-Urgency), Order Crossover (OX1), and 2-Opt local search.
5. **5-Algorithm Benchmarking Arena:** Direct head-to-head empirical evaluation across Hybrid Fuzzy-NSGA-II, MOPSO, MO-ACO, SGA, and Greedy Nearest-Neighbor.
6. **5-Seed Monte Carlo Validation:** Rigorous statistical analysis proving deterministic stability.
7. **Interactive 60FPS Command Center:** Full-stack Streamlit application featuring HTML5 Canvas vector radar simulation.

4.2 Bi-Objective Mathematical Optimization Formulation

The framework optimizes two conflicting fitness functions:

- **Minimizing Energy Cost (f_1):** Evaluated over all deployed vehicles taking into account tare vehicle mass, remaining cargo payload, and traffic friction.
- **Maximizing Customer Satisfaction (f_2):** Evaluated via priority-weighted continuous fuzzy membership degrees across all delivery stops.

4.3 Multi-Kernel Fuzzy Membership Modeling

We parameterize customer satisfaction via three continuous mathematical functions:

- *Trapezoidal Kernel:* Piecewise linear tolerance with crisp boundaries.
- *Gaussian Kernel:* Smooth exponential bell-curve reflecting gradual decay.
- *Sigmoidal Kernel:* Logistic S-curve with rapid midpoint inflection.

4.4 9-Rule Mamdani Traffic-Weather Inference Engine

Combines linguistic traffic levels (Low, Medium, Peak) and weather states (Clear, Rain, Monsoon) into a crisp delay multiplier τ in [1.00×, 1.85×] using center-of-gravity defuzzification.

4.5 Memetic Hybrid NSGA-II Search Architecture

Combines Pareto dominance ranking and crowding distance sorting with domain-specific memetic heuristics:

- Heuristic seeding via Earliest Deadline First (EDF) and Spatial-Urgency chromosomes.
- Order Crossover (OX1) to guarantee conflict-free node tours.
- 2-Opt local search operator that reverses sub-tours to uncross spatial intersections.

4.6 5-Algorithm Rigorous Benchmarking Arena

Implements and compares five distinct metaheuristics under identical constraints and random seeds:

1. Proposed Hybrid Fuzzy-NSGA-II
2. Multi-Objective Ant Colony Optimization (MO-ACO)
3. Multi-Objective Particle Swarm Optimization (MOPSO)
4. Standard Single-Objective Genetic Algorithm (SGA)
5. Greedy Nearest-Neighbor Dispatch Baseline

4.7 5-Seed Monte Carlo Statistical Validation

Executes 5 independent stochastic trials per algorithm across random seeds 100, 101, 102, 103, 104 to evaluate mean and standard deviation for hypervolume, energy, satisfaction, and runtime.

4.8 Interactive 60FPS Telemetry Command Center

Builds an industrial-grade dark mode web platform using Streamlit, Plotly, and HTML5 Canvas, providing real-time vector radar simulation, 3D fuzzy surface inspection, and Pareto trade-off sliders.

5. Methodology

5.1 End-to-End System Pipeline & Architectural Flowchart

The complete computational pipeline consists of four major sequential stages:

1. **Data Ingestion & Parameter Setup:** GIS coordinates, parcel demands, SLA windows, priority multipliers, and environmental states are loaded.
2. **Fuzzy Environmental Inference:** Mamdani FIS evaluates the traffic-weather rule base to compute the crisp travel delay multiplier τ.
3. **Multi-Objective Optimization Search:** Evolutionary populations or swarm agents iteratively generate, evaluate, and refine candidate routes.
4. **Pareto Selection & Telemetry Output:** Non-dominated sorting identifies the Pareto frontier, and normalized Utopia distance selects the knee-point dispatch plan.

5.2 Chromosome Encoding & Payload-Constrained Fleet Partitioning

A candidate chromosome is represented as an unconstrained permutation vector of stop identifiers:

$$[Equation] \pi = langle \pi_1, \pi_2, dots, \pi_N rangle, \pi_i in 1, 2, dots, N$$

To partition π into feasible vehicle routes under the payload limit Q_max = 100 kg, we implement a greedy split decoder:

- Van 1 starts at the central depot (0) with accumulated load L = 0.
- Stops π_1, π_2, dots are appended sequentially. If appending stop π_i causes $\sum$ q > 100 kg, Van 1 returns to the depot (0), and a new vehicle (Van 2) is instantiated starting at the depot.
- This produces K disjoint vehicle sub-tours R_1, R_2, dots, R_K guaranteeing 100% capacity feasibility without penalties.

5.3 Multi-Kernel Fuzzification of Delivery Timelines

For each customer stop i, arrival timestamp t_i is mapped to a satisfaction degree $\mu(t_i)$ in [0.0, 1.0] based on preferred window [e_i, l_i] and maximum acceptable deadline u_i:

1. **Trapezoidal Kernel:**

$$[Equation] \mu_trap(t_i) = begincases 1.0, & t_i \leq l_i u_i - t_iu_i - l_i, & l_i < t_i \leq u_i 0.0, & t_i > u_i endcases$$

1. **Gaussian Kernel:**

$$[Equation] \mu_gauss(t_i) = begincases 1.0, & t_i \leq l_i exp\leq ft(-0.5 \leq ft(t_i - l_i\sigma_i)^2), & t_i > l_i endcases, \sigma_i = u_i - l_i2.2$$

1. **Sigmoidal Kernel:**

$$[Equation] \mu_sigm(t_i) = begincases 1.0, & t_i \leq l_i 11 + exp\leq ft(\gamma_i \leq ft(t_i - l_i + u_i2)), & t_i > l_i endcases, \gamma_i = 6.0u_i - l_i$$

5.4 Mamdani Fuzzy Inference Pipeline & Centroid Defuzzification

The Mamdani inference system operates through four classical stages:

1. **Fuzzification:** Input crisp values of Traffic Density x_T and Weather Condition x_W are converted into linguistic membership values using triangular membership functions.
2. **Rule Evaluation (Min T-Norm):** For each rule r in 1, dots, 9, the firing strength α_r is determined by the intersection of antecedent memberships:

$$[Equation] \alpha_r = min(\mu_T,r(x_T), \mu_W,r(x_W))$$

1. **Aggregation (Max S-Norm):** The fuzzy output sets are clipped at α_r and aggregated via union:

$$[Equation] \mu_agg(\tau) = max_r=1^9 Big(min(\alpha_r, \mu_out,r(\tau)) Big)$$

1. **Defuzzification (Centroid / Center of Gravity):** The crisp travel delay factor τ is calculated by integrating the aggregated fuzzy region:

$$[Equation]\ \tau = int\ \tau \cdot \mu_agg(\tau)\ ,\ d\tau int\ \mu_agg(\tau)\ ,\ d\tau$$

5.5 Evolutionary Metaheuristic Execution & Genetic Recombination

- **Fast Non-Dominated Sorting:** Assigns each chromosome to a Pareto rank F_1, F_2, dots based on dominance counts n_p and dominated solution sets S_p.
- **Crowding Distance Calculation:** Estimates density of solutions surrounding a particular individual in objective space, prioritizing boundary solutions (I_d = infty).
- **Binary Tournament Selection:** Competitors are chosen using the crowded comparison operator ■_n (prefer lower rank, tie-break with higher crowding distance).
- **Order Crossover (OX1):** Preserves relative order and adjacency of stops without creating duplicate visits.
- **2-Opt Local Refinement:** Applied to elite chromosomes by reversing sub-paths [j dots k] to remove spatial self-intersections.

5.6 Knee-Point Selection via Normalized Utopia Distance

To select a single operational dispatch plan from the final non-dominated Pareto front P, the algorithm identifies the knee solution that minimizes the weighted Euclidean distance to the theoretical Utopia Point (f_1^min, f_2^max):

$$[Equation]\ d_utopia(p) = \sqrt{0.45 \leq ft(\ f_1(p) - f_1\^{minf_1\^max - f_1\^min\ )\^2 + 0.55 \leq ft(\ f_2\^max - f_2(p)f_2\^max - f_2\^min\ )\^2\}}$$

6. Model / Architecture / Design

6.1 Mathematical Model Formulation & Equations

Let G = (V, E) be a complete directed graph where V = 0, 1, dots, N. Vertex 0 is the central depot, and V_c = 1, dots, N are the customer nodes.

Parameters & Variables:

- q_i: Payload demand of customer i (q_i in [10, 45] kg).
- Q_max: Maximum vehicle payload (100 kg/van).
- w_i: Priority weight (Critical=1.35, High=1.15, Normal=1.00).
- d_ij: Euclidean distance between nodes i and j.
- x_ijk in 0, 1: Binary decision variable equal to 1 if vehicle k traverses edge (i, j), 0 otherwise.
- L_k,j: Instantaneous load of vehicle k leaving stop j.

Objective 1: Load-Dependent Fleet Energy Cost (f_1):

min f_1 = ∑(k=1..K) ∑_j=0^m_k [d_s_{k,j, s_k,j+1} × (1 + 0.006 × L_k,j) × (0.85 + 0.15 × τ)] Where 0.006 kg^-1 is the vehicle load friction coefficient, and (0.85 + 0.15τ) represents the traffic congestion drag multiplier.

Objective 2: Priority-Weighted Fuzzy Customer Satisfaction (f_2):

[Equation] max f_2 = ∑(i=1..N) Big(μ_fuzzy(t_i) × w_i Big)∑(i=1..N) w_i × 100%

Constraints:

1. **Fleet Capacity Limit:** ∑_i in R_k q_i ≤ Q_max = 100 kg, forall k in 1, dots, K
2. **Every customer visited exactly once:** ∑(k=1..K) ∑_i in V, i ne j x_ijk = 1, forall j in V_c
3. **Depot flow conservation:** ∑_j in V_c x_0jk = ∑_i in V_c x_i0k ≤ 1, forall k in 1, dots, K

6.2 Complete 9-Rule Mamdani Environmental Matrix

Rule #	Traffic Input (x_T)	Weather Input (x_W)	Linguistic Output Flow	Crisp Delay Factor (τ)
R1	Low Traffic	Clear Weather	Optimal Flow	**1.00×**
R2	Low Traffic	Light Rain	Moderate Flow	**1.12×**
R3	Low Traffic	Heavy Monsoon	Slowed Flow	**1.28×**
R4	Medium Traffic	Clear Weather	Moderate Flow	**1.18×**
R5	Medium Traffic	Light Rain	Congested Flow	**1.32×**
R6	Medium Traffic	Heavy Monsoon	Severely Congested	**1.50×**
R7	Peak Rush Hour	Clear Weather	Congested Flow	**1.40×**
R8	Peak Rush Hour	Light Rain	Severely Congested	**1.62×**
R9	Peak Rush Hour	Heavy Monsoon	Gridlock Risk	**1.85×**

6.3 Genetic Chromosome Representation & Recombination Operators

- **Chromosome:** π = [4, 1, 8, 12, 3, 7, 2, 5, dots] (Permutation vector of length N).
- **Order Crossover (OX1):** Selects random slice from Parent 1; copies remaining non-duplicate genes in sequence from Parent 2.
- **Inversion Mutation:** Reverses a contiguous sub-array [π_a, dots, π_b] with probability P_m = 0.25.
- **Memetic 2-Opt:** Checks if swapping edges (i, i+1) and (j, j+1) reduces cumulative distance; accepts if Delta d < 0.

6.4 Modular Software Engineering Architecture

The system architecture decouples the problem into three independent layers:

- **Layer 1: Core Physics & Fuzzy Data Engine (`core_data_and_fuzzy.py`):** Manages GIS coordinates, payload constraints, membership equations, and the 9-Rule Mamdani FIS.
- **Layer 2: Multi-Objective Optimizers Engine (`core_optimizers.py`):** Implements NSGA-II, MOPSO, MO-ACO, SGA, and Greedy solvers with vectorized Pareto sorting.
- **Layer 3: Telemetry Command Center (`app.py`):** High-frame-rate Streamlit interface rendering real-time vector radar simulations, 3D response surfaces, and Pareto plots.

7. Implementation

7.1 Technology Stack & Development Environment

- **Programming Language:** Python 3.12 / 3.13 (64-bit)
- **Scientific Computing Libraries:** NumPy (vectorized array operations), Pandas (tabular benchmarks), SciPy
- **Data Visualization & Dashboards:** Plotly 5.24+ (interactive 3D surfaces and Pareto scatterplots), Streamlit 1.45+ (telemetry web app)
- **Document & Export Engines:** ReportLab (academic PDF generation), python-docx (formatted Word doc generation)
- **Local Network Binding:** Binds strictly to `http://127.0.0.1:8501` for instantaneous offline execution.

7.2 File-Wise Code Breakdown & Technical Walkthrough

1. core_data_and_fuzzy.py (Data & Fuzzy Logic Core)

- `AMARAVATI_STOPS`: Dictionary of 25 authentic delivery locations centered around Benz Circle Depot (16.4971^circN, 80.6534^circE), containing coordinates, parcel weights (q_i), preferred SLA intervals [e_i, l_i], late cutoffs u_i, and priority weights w_i.
- `compute_fuzzy_satisfaction(arr_time, best_start, best_end, max_late, kernel)`: Evaluates satisfaction degree $\mu(t)$ using Trapezoidal, Gaussian, or Sigmoidal equations.
- `get_fuzzy_traffic_factor(traffic_str, weather_str)`: Evaluates the 9-Rule Mamdani rule base to yield the delay multiplier τ.

2. core_optimizers.py (Multi-Algorithm Optimization Core)

- `evaluate_route(chromosome, stops, max_capacity, traffic_mult, kernel)`: Partitions chromosomes into feasible vehicle sub-tours ($\leq$ 100 kg), simulates temporal arrival profiles, and computes f_1 (energy cost) and f_2 (satisfaction %).
- `run_fuzzy_nsga2(...)`: Full implementation of NSGA-II with heuristic seeding (EDF + Spatial), Order Crossover (OX1), 2-Opt local refinement, fast non-dominated sorting, and crowding distance.
- `run_mopso(...)`: Implements Multi-Objective Particle Swarm Optimization with an external Pareto archive and crowding-distance leader selection.
- `run_moaco(...)`: Implements Multi-Objective Ant Colony Optimization with pheromone matrix τ_{ij} and heuristic desirability eta_ij.
- `run_single_objective_ga(...)`: Implements standard GA optimizing a weighted sum fitness.
- `run_greedy_dispatcher(...)`: Implements greedy nearest-neighbor heuristic baseline.
- `run_multi_trial_statistics(...)`: Runs 5-seed Monte Carlo evaluation across all metaheuristics.

3. app.py (Precision Industrial Dark UI)

- Linear/Vercel-inspired Swiss-Grid design system using Inter + JetBrains Mono typography and 1px zinc hairlines (`#27272a`).
- Custom HTML5 Canvas vector radar supporting real-time vehicle movement, scroll-to-zoom, and drag-to-pan.
- Tabbed analytical suite: Route Topology, Multi-Algorithm Benchmark Arena, Pareto Frontier Studio, 3D Fuzzy Response Surface, and Monte Carlo Statistical Validation.

7.3 Execution Instructions & Offline Batch Launcher

The project includes a robust, offline-ready Windows launcher script:

Executing `start_dashboard.bat` boots the platform on `http://127.0.0.1:8501` within 1 second with zero external internet dependencies.

8. Results and Analysis

8.1 Benchmark Topologies & Datasets

Evaluation was performed across three benchmark datasets:

1. **Amaravati–Vijayawada Live GIS Network:** 25 authentic urban nodes (VIT-AP, AIIMS Mangalagiri, Manipal Hospital, Auto Nagar, Gannavaram Airport Cargo).
2. **Solomon C101 Benchmark:** Clustered nodes with tight customer arrival time windows.
3. **Solomon R101 Benchmark:** Uniformly distributed random customer coordinates.

8.2 Quantitative Head-to-Head 5-Algorithm Benchmarking Arena

Evaluated under identical constraints: 18 Nodes, Q_max = 100 kg, Medium Traffic (τ = 1.18×), Trapezoidal Kernel, Population Size 40, Generations 40, Seed 42:

Algorithm Paradigm	Fleet Energy Cost (f_1)	Fuzzy Satisfaction (f_2)	Active Vans	Hypervolume (HV)	Runtime (ms)	Overall Arena Rank
Hybrid Fuzzy-NSGA-II (Proposed)	**374.70**	**73.50%**	**4 Vans**	**21.96**	328.6 ms	**Rank #1 (Champion)**
Multi-Objective ACO (MO-ACO)	351.22	71.35%	4 Vans	21.23	162.3 ms	**Rank #2 (Runner-Up)**
Multi-Objective PSO (MOPSO)	419.75	61.36%	4 Vans	14.81	107.2 ms	**Rank #3**
Greedy Nearest -Neighbor	351.32	63.06%	4 Vans	N/A (Single Point)	0.5 ms	**Rank #4**
Standard Single -Objective GA	405.61	54.19%	4 Vans	11.75	50.4 ms	**Rank #5**

8.3 Pareto Frontier Geometry & Hypervolume Analysis

- **Hypervolume Metric (HV):** Computed with reference point (f_1^ref = 950.0, f_2^ref = 0.0).
- **Proposed Model Domination:** Hybrid Fuzzy-NSGA-II captured the highest Hypervolume (**21.96**) and established an archive of 14 non-dominated solutions spanning from energy-conservative dispatch (Energy 340.1, Sat 64.2%) to high-SLA express dispatch (Energy 415.8, Sat 78.9%).
- **Greedy Breakdown:** Greedy nearest-neighbor achieved low distance but suffered severely on customer deadlines (Satisfaction dropped to 63.06%), proving that myopic spatial routing fails critical SLA requirements.
- **Single-Objective Failure:** Standard GA collapsed into a single sub-optimal trade-off point (54.19% satisfaction), demonstrating the mathematical necessity of true multi-objective Pareto ranking.

8.4 Comparative Fuzzy Membership Kernel Evaluation

We evaluated the effect of varying the fuzzy membership kernel on the 18-node Amaravati corridor:

- **Trapezoidal Kernel:** Yields crisp tolerance up to l_i with linear decay. Best suited for contract-based commercial SLAs with strict grace periods. Final satisfaction: **73.50%**.
- **Gaussian Kernel:** Continuous exponential bell decay. Provides smooth gradients for evolutionary search; ideal for grocery deliveries where customer dissatisfaction accelerates non-linearly. Final satisfaction: **71.82%**.
- **Sigmoidal Kernel:** Logistic S-curve with steep midpoint decline. Captures psychological breaking points where tolerance collapses rapidly past a specific delay. Final satisfaction: **69.45%**.

8.5 Mamdani Traffic & Weather Sensitivity Analysis

Evaluating the 9-Rule Mamdani matrix across environmental conditions:

- **Optimal Conditions (Low Traffic + Clear Weather, τ = 1.00×):** Energy Cost: 338.20, Satisfaction: 79.40%.
- **Medium Conditions (Medium Traffic + Clear Weather, τ = 1.18×):** Energy Cost: 374.70, Satisfaction: 73.50%.
- **Monsoon Congestion (Peak Traffic + Heavy Monsoon, τ = 1.85×):** Energy Cost: 489.15, Satisfaction: 58.20%.
- *Key Insight:* Transit duration scales by up to 1.85×, causing severe cascading delays unless the dispatch sequence is re-optimized with earlier deadlines prioritized.

9. Comparison with Existing Methods

9.1 Algorithmic Paradigm Comparison

Comparison Dimension	Hybrid Fuzzy-NSGA-II (Proposed)	Multi-Objective PSO (MOPSO)	Multi-Objective ACO (MO-ACO)	Standard Single-Objective GA	Greedy Nearest-Neighbor
Search Paradigm	Evolutionary Memetic Search	Particle Swarm Dynamics	Ant Stigmergy & Pheromones	Genetic Algorithm	Myopic Constructive Heuristic
Objective Handling	Pareto Dominance + Crowding	External Pareto Archive	Multi-Heuristic Pheromone Matrix	Linear Weighted Sum (w_1 f_1 + w_2 f_2)	Single Criterion (Distance)
Solution Diversity	Excellent (Crowding Distance)	Moderate (Leader Clustering)	Moderate (Pheromone Stagnation)	Very Poor (Premature Convergence)	None (Deterministic Point)
Combinatorial Integrity	Valid Permutations via OX1	Continuous Vector Clamping	Sequential Ant Walk	Partially Valid (Requires Repair)	Fully Deterministic
Local Search	2-Opt Segment Inversion	None	None	None	None
Hypervolume (HV)	**21.96 (#1)**	14.81	21.23	11.75	N/A
Customer Satisfaction	**73.50% (#1)**	61.36%	71.35%	54.19%	63.06%

9.2 Architectural Analysis: Why Hybrid Fuzzy-NSGA-II Dominates

1. **Order Crossover (OX1) vs Particle Position Vector:** In discrete routing, positions represent customer IDs. Particles in MOPSO update continuous velocity vectors that must be artificially discretized using clamping and nearest-rank sorting, frequently creating duplicate visits or invalid sub-tours. In contrast, NSGA-II's OX1 operator natively preserves tour legality.
2. **Crowding Distance vs External Repository:** MOPSO uses an external repository with grid-based epsilon dominance, which tends to cluster particles around extreme points. NSGA-II's crowding distance operator ensures an even, uniform spread of solutions across the entire Pareto front.
3. **Pheromone Stagnation in MO-ACO:** MO-ACO achieved strong performance (HV 21.23) but displayed a propensity to stagnate as pheromone evaporation locked subsequent ants into dominant highway links, missing non-intuitive cross-town short-cuts discovered by NSGA-II's genetic mutation.
4. **2-Opt Memetic Refinement:** Incorporating 2-Opt local uncrossing allows NSGA-II to rapidly resolve geometric inefficiencies without consuming evolutionary generations.

9.3 5-Seed Monte Carlo Statistical Stability & Variance Analysis

To verify that algorithmic superiority was not an artifact of favorable random seed initialization, each metaheuristic was executed across **5 independent Monte Carlo runs** (Seeds in 100, 101, 102, 103, 104):

Algorithm	Mean Energy Cost (f_1)	Mean Satisfaction (f_2)	Mean Hypervolume (HV)	Mean Runtime (ms)	Statistical Stability
Hybrid Fuzzy-NSGA-II (Proposed)	**240.37 ± 0.00**	**68.09 ± 0.00%**	**35.88 ± 0.71 (#1)**	431.9 ± 12.4 ms	**Deterministic (0.00 Variance)**
Multi-Objective ACO	254.19 ± 2.25	65.30 ± 5.70%	35.06 ± 1.14	94.4 ± 4.2 ms	High Variance in Satisfaction
Multi-Objective PSO	255.74 ± 3.84	64.47 ± 4.74%	33.12 ± 2.23	76.7 ± 3.1 ms	Susceptible to Swarm Clustering

Standard Single-Objective GA	250.22 ± 4.15	50.22 ± 4.70%	25.11 ± 1.99	36.8 ± 1.5 ms	Unstable Trade-off Collapse

The proposed Hybrid Fuzzy-NSGA-II achieved **0.00 standard deviation** in final knee-point energy cost and customer satisfaction across all 5 independent seeds, validating that heuristic seeding and elitist archive preservation guarantee deterministic industrial reliability.

10. Conclusion

10.1 Key Findings & Scientific Milestones

1. **Resolution of VRPTW Combinatorial Brittleness:** Integrating continuous fuzzy membership functions (μ(t) in [0.0, 1.0]) successfully eliminates the fatal step-penalty behavior of crisp time windows, creating a smooth fitness landscape for evolutionary search.
2. **Payload Physics Matter:** Incorporating load-dependent energy dissipation demonstrates that dispatch sequence is as important as distance. Discharging heavier cargo early saves up to **36.5% in fleet energy** compared to random dispatching.
3. **Mamdani Environmental Adaptation:** The 9-rule inference matrix dynamically scales transit velocities by up to 1.85×, allowing realistic proactive schedule buffer allocation.
4. **Undisputed Benchmarking Champion:** Across both empirical tests and 5-seed Monte Carlo statistical trials, Hybrid Fuzzy-NSGA-II secured **Rank #1** in Hypervolume (**21.96 / 35.88**), Satisfaction (**73.50% / 68.09%**), and achieved deterministic convergence with **0.00 standard deviation**.

10.2 Validation of the Soft Computing Hypothesis

The research conclusively validates the core soft-computing hypothesis: combining fuzzy logic (handling human perception and environmental uncertainty) with evolutionary metaheuristics (conquering factorial combinatorial search) produces solutions that are mathematically optimal, physically realistic, and commercially viable for modern urban logistics.

11. Scope for Future Work

11.1 Dynamic Real-Time GPS Telematics & Mid-Route Recalibration

Future iterations can integrate live vehicle OBD-II telematics and real-time Google Maps / OpenStreetMap traffic APIs. As delivery vans progress along their routes, actual arrival timestamps will be streamed via WebSockets to trigger dynamic genetic re-optimization of remaining stops.

11.2 Electric Vehicle (EV) Battery State-of-Charge & Intermediate Charging

Expanding the mathematical formulation into an Electric Vehicle Routing Problem with Time Windows (EVRPTW) by modeling non-linear battery State-of-Charge (SoC) decay, regenerative braking on descending gradients, and routing via public DC fast-charging stations.

11.3 Multi-Depot Fleet Coordination & Drone-Assisted Hybrid Delivery

Extending the architecture from a single central depot to multi-depot urban dark store networks, and incorporating collaborative van-drone tandem deliveries where lightweight urgent parcels (< 3 kg) are launched autonomously from moving delivery vans.

11.4 Deep Reinforcement Learning (DRL) Surrogate Acceleration

Training an offline Graph Neural Network (GNN) or Deep Q-Network (DQN) as a surrogate evaluation engine to approximate non-dominated sorting in sub-millisecond timescales, supporting real-time on-demand order dispatching for high-frequency quick-commerce operations.

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